

optimize your hardware

an intelligent decision matrix, automated workload profiling, and power analytics engine built to eliminate hardware bottlenecks.

+100%
rule-based precision

WORKLOAD PROFILING

TDP ANALYTICS

RULE-BASED ENGINE

SCROLL

7
components analyzed

SYSTEM SPECIFICATIONS & MODULES



SPECS 01 | PROCESSOR

CPU Architecture & Cores

Optimal core counts, clock speeds, L3 cache sizing, and AVX-512 instructions calculated for high-parallel workloads.



SPECS 02 | GRAPHICS

GPU & VRAM Bottlenecking

Rule-based mapping for VRAM buffer requirements, CUDA/Tensor core scaling, and ray-tracing pipeline stability.



SPECS 03 | MEMORY & NVME

RAM Bandwidth & Speed

Calculates dual-channel DDR5 bus bandwidth, NVMe Gen5 throughput, and random I/O latency targets.



SPECS 04 | POWER & THERMAL

TDP Wattage & Thermals

Real-time estimation of total system power draw, transient spike margins, liquid cooling radiator sizing, and fan noise profiles.

3. INTERACTIVE REQUIREMENT MATRIX

Configure Your Target Workload & Upgrades

Select environment, intended usage, and budget. Tweak live sliders to inspect upgrade scaling like a custom PC builder.

ENVIRONMENT TYPE

Basic Home / Student PC (Entry)

TARGET PRIMARY USE

Basic Web Browsing & Online Classes

TARGET BUDGET BRACKET

Budget: ₹25,000

DYNAMIC HARDWARE UPGRADE SLIDERS

RAM Capacity:

4 GB DDR4

Storage Speed & Capacity:

256 GB NVMe SSD

GPU Compute Tier:

Integrated Graphics (iGPU)

PC BUILDER REAL-TIME SPECIFICATIONS

Estimated Power Draw (TDP):

148.75 Watts

Recommended PSU Wattage:

450W Power Supply

Memory Bandwidth:

25.6 GB/s (DDR4)

Storage Throughput:

256 GB NVMe SSD

Thermal / Fan Noise:

22 dB (Ultra Quiet)

Bottleneck Ratio:

6.4% (Optimal Balance)

Est. Total Build Cost:

₹21,370

4K Gaming FPS

31 FPS

AI Token Rate

6 tok/s

3D Render Speed

1.0x Speed

4. OPTIMAL BUILD RECOMMENDATION

Recommended Hardware & Component Gallery

Based on your selected workload matrix and slider preferences, here are the non-bottlenecked component recommendations from our CSV hardware database.

BEST VALUE STUDENT

PROCESSOR MODEL

SELECT CPU:

Intel Core i3-12100 (Best Value Student)

Intel Core i3-12100

4 Cores / 8 Threads, 4.3GHz Turbo, Intel UHD 730 iGPU

Power Draw: 60W TDP Power Draw

₹7,990

INTEGRATED iGPU

GRAPHICS ACCELERATION

Intel / AMD Integrated UHD 730 / Vega 8 iGPU

Shared System Memory, DirectX 12, 4K Display Output

CUDA Cores: 15W TDP Power Draw

Included with CPU

ENTRY COMPACT

MOTHERBOARD

ASUS PRIME H610M-K D4

LGA1700 mATX, PCIe 4.0 NVMe, USB 3.2 Gen 1, HDMI

VRM Phases: H610 Chipset

₹6,490

SYSTEM MEMORY

4GB Crucial DDR4-3200MHz

Single-Channel Value Kit

Latency: CL22 @ 3200MHz

₹1,120

HIGH-SPEED STORAGE

Crucial P3 256GB M.2 PCIe 3.0 NVMe

256 GB NVMe SSD

Endurance: 2,400 TB/s Read Speed

₹2,190

CHASSIS / CASE

Ant Esports SI24 Micro-ATX Cabinet

Compact mATX Tower, Front USB 3.0, Steel Side Panel

AirFlow: Micro-ATX Form Factor

₹1,890

BASIC POWER

450W Ant Esports VS450L

450W Continuous Output, 120mm Silent Fan

Efficiency: 450W Standard Power Supply

₹1,690

5. PERFORMANCE & DIAGNOSTIC SYSTEM

Generate Final Diagnostic Hardware Report

Provides execution time estimates, TDP calculations, and architecture evaluation scores in downloadable summaries across all 7 components.

100%

Rule-based automated hardware mapping

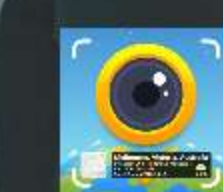
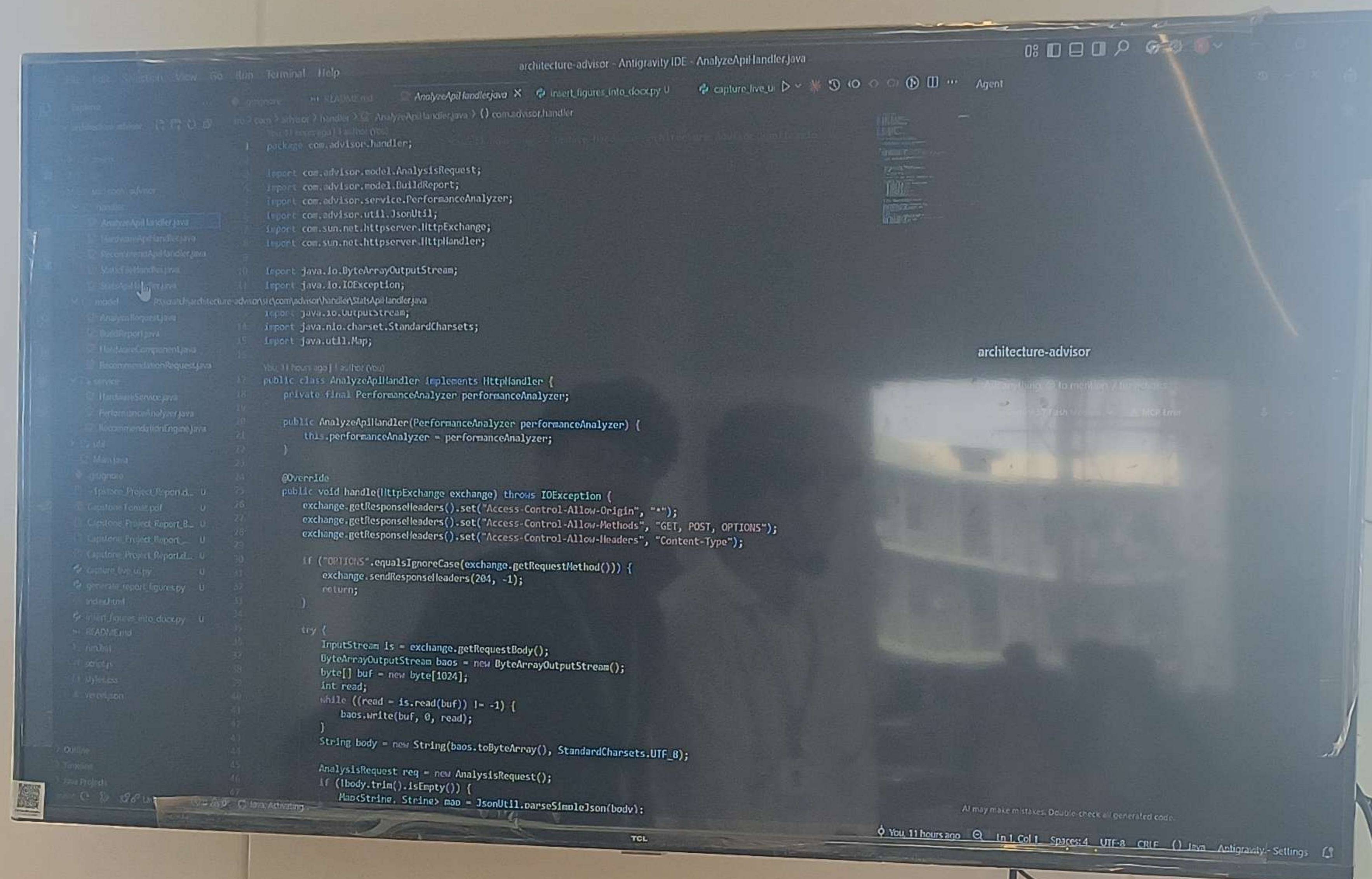
7 Modules

Requirement profiling to full export

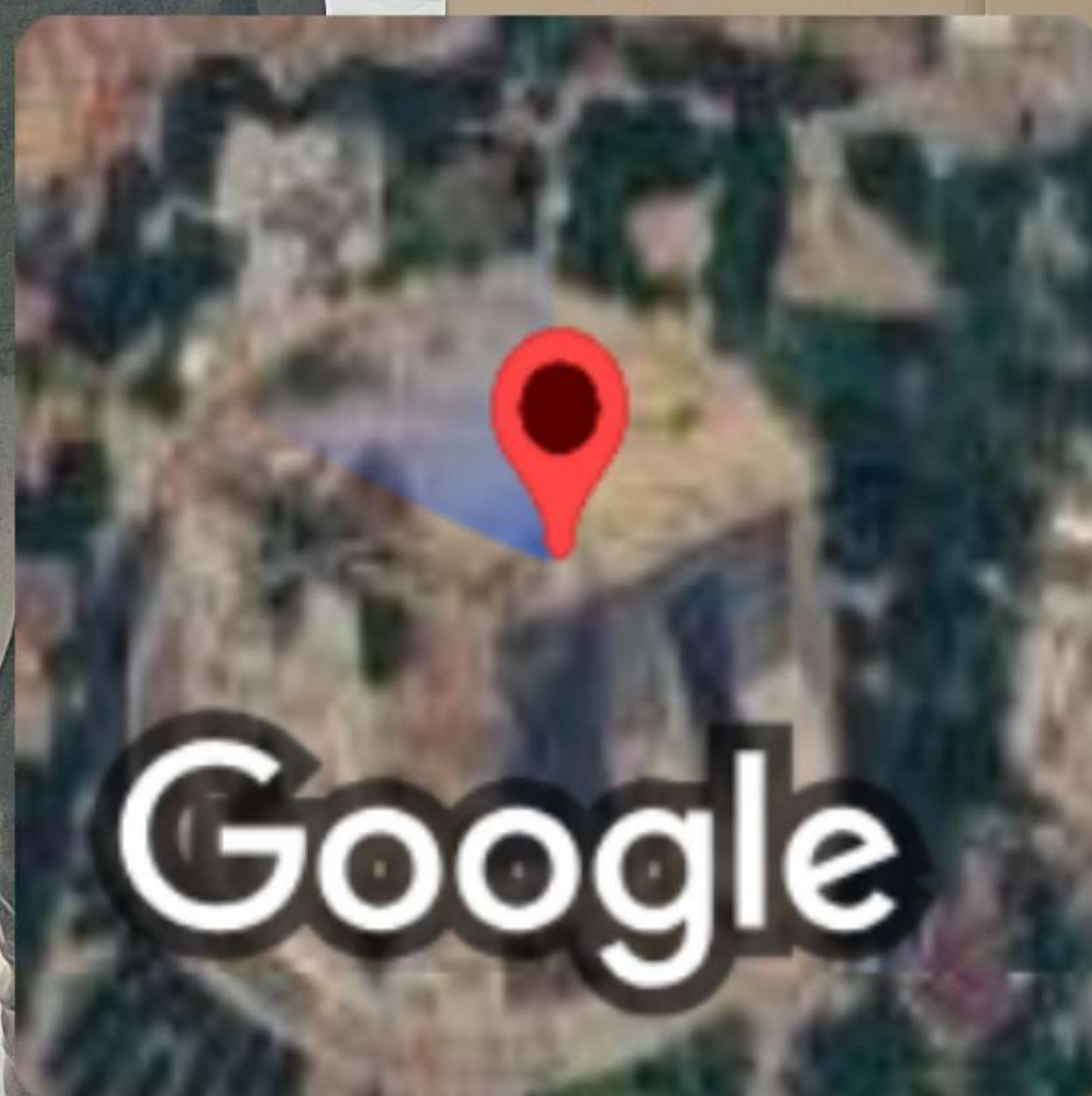
5 Profiles

Workload environments supported

GENERATE DIAGNOSTIC REPORT



GPS Map Camera



Mevalurkuppam, Tamil Nadu, India



22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.026744° Long 80.016757°

Wednesday, 02/09/2026 08:25 AM GMT +05:30